A CHANNEL CHANGE QUADRUPLES MISSED SPOOFS WHILE THE FALSE-ALARM RATE READS BETTER THAN TARGET

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ABSTRACT

We calibrate a speech-deepfake detector at 5% FPR on speech carried over a real PSTN connection and deploy it on the same recordings retransmitted over G.722 VoIP. The transported threshold realizes **0.47% FPR** while missing **74% of spoofs, versus 19% at the deployment-calibrated threshold**. Across 108 deployment cells, 58 miss target by more than 2×; 34 are conservative failures that pass an additive ±5-point audit. The failure persists on ASVspoof 5 when calibration and deployment share neither recordings nor speakers: 187 of 264 ordered pairs (71%) miss by more than 2× across two detectors. Spoof-side cost there is measurable for SSL-AASIST only, because AASIST fails our viability gate. A target-channel bona-fide order statistic restores *marginal expected* FPR control: at $N{=}500$, mean realized FPR is 4.94–5.03%, with at most a 0.8-point FNR premium on viable cells. This is not a per-deployment guarantee. Exchangeability gives marginal rank validity. The conditional-FPR Beta law requires iid continuous sampling from a fixed distribution; speaker clustering widens dispersion on the flagship channel. The tested label-free corrections do not restore control, and no tested unlabeled monitor survives an attack-prevalence shift. Threshold transport can therefore fail quietly. Target-channel calibration restores marginal control, but its sampling unit and dispersion must be reported.

Index Terms— Anti-spoofing, audio deepfake detection, threshold calibration, conformal prediction, domain shift

1. INTRODUCTION

Audio anti-spoofing studies usually report performance at the EER threshold, an oracle operating point selected with labels from both evaluation classes. A deployed service has no such oracle: it must choose a threshold in advance and accept the resulting false-positive rate. This difference can hide severe deployment failures. A threshold set to 5% FPR on ASVspoof 2019 LA dev [1] realizes 30–98.5% FPR across 21LA/21DF [2], In-the-Wild [3], and BRSpeech-DF [4] for a strong open-source detector [5]; for XLS-R+SLS [6], it reaches 99.5%. These results agree with the transfer failures in [7] and the bona-fide-driven cross-testing errors in [8]. Channel is a major source of variation: [9] finds that presentation can matter more than model scale, and a recent challenge targets this problem directly [10]. Language introduces a complementary resource asymmetry. LRLspoof [11] transfers an EER threshold across 66 languages but, without target-domain bona fide, can report only spoof rejection. Our deployment regime assumes the converse resource: target bona fide is available, but target spoof labels are not. We therefore evaluate the FPR side.

We study the deployment behavior of a standard statistical repair. Given N labeled target-domain *bona-fide* recordings, but no costly spoof labels, the $\lfloor (N{+}1)\alpha \rfloor$-th order statistic has marginal rank validity under exchangeability without ties. Under iid continuous sampling from a fixed score distribution, the conditional FPR across calibration draws additionally follows an exact finite-sample Beta law [12, 13]. Conservative Neyman–Pearson variants instead provide high-probability upper bounds [12, 14]. Neither the statistic nor bona-fide-only audio calibration is new [15, 16]. Our contribution is empirical: we measure transport across channels and corpora, quantify its operational cost, replicate the failure on disjoint data, and compare finite-sample policies at matched labeling budgets. We ask whether target-channel calibration leaves a *usable* operating point, what alternative policies buy with the same labels, and where the sampling assumptions fail in practice.

The failure is operationally quiet: on the cells that matter the transported threshold collapses *conservative*, so observable FPR reads on target or better while missed spoofs multiply. Spoof-side cost needs the unavailable class; low FPR alone cannot distinguish control from collapse.

We make three contributions (details in §§4–5). (1) The measurement covers 108 deployment cells: two detectors × (42 ordered pairs of seven 21LA channel conditions + 12 ordered pairs of four corpora). We report both the loss of false-alarm control and the spoof-side cost relative to each deployment's oracle. (2) We re-audit the bona-fide quantile against parametric and unlabeled alternatives at matched resource. (3) We report negative results: the tested label-free heuristics do not restore control, and no tested unlabeled drift monitor survives an attack-prevalence shift. We claim neither the threshold rule nor the practice of auditing both error rates at a fixed operating point; both are prior art [12, 13, 17, 18]. Our two detectors lose control by different amounts, but they differ in both front-end and static accuracy, so n=2 cannot separate the explanations.

2. RELATED WORK

Order-statistic calibration is established: Neyman–Pearson methods control type-I error through class-0 order statistics [12, 14], negative-only conformal calibration provides marginal guarantees [13], and related FPR bounds have been studied for generated *text* [16]. In audio, CA-SOADD calibrates a bona-fide-only α-quantile [15], Reality Check fixes thresholds before evaluation [19], [7] audits threshold transfer and seven unlabeled corrections, and [8] documents bona-fide-driven cross-testing errors. Relative to that audit's one detector and two target corpora, we add a 108-cell fixed-FPR map that includes real 21LA transmissions, a finite-N target-bona-fide comparison, and a 264-pair recording- and speaker-disjoint replication. We do not claim the first quantile rule or the first threshold audit. An industry report describes the same failure without reproducible figures [20]. Existing real-channel benchmarks still use oracle-selected metrics: 21LA reports minimum t-DCF, while RTCFake reports EER over 600 hours transmitted through seven consumer platforms [21].

Calibration is also well established in speech evaluation. Challenges audit fixed-cost operating points through t-DCF/a-DCF [17]; speaker recognition has long used C_{llr} and application-independent calibration measures [22]; and anti-spoofing has introduced per-artifact evidence scores [23]. These methods use both classes, which are unavailable in our setting. Le Roux et al. [24] calibrate a POI-conditional detector from $N \in \{3..50\}$ recordings using heuristic rules. We instead study general countermeasure scores, specify a finite-sample estimand, and compare policies at the same labeling budget.

Prior work detects deployment drift and then retrains the model [25]; we hold the detector fixed and ask whether its threshold remains controlled. Distributional distance alone does not certify an FPR band, and none of our monitors survives an attack-prevalence shift. In a separate pre-registered test, adding 5% spoofs to the calibration cohort pushes realized FPR below target on seven of eight cells (0.99–4.99%); the eighth, overlap-dominated cell is nearly unchanged at 5.11%. Non-exchangeable conformal theory does not determine the direction of error [26]. Covariate-shift weighting [27], outcome-adaptive conformal inference [28], and contamination-robust methods [29] motivate distinct remedies, but we test only the score-weighting heuristic and oracle-label adaptive ceiling defined in §4. The negative result is therefore not attributed to those constructions. Closest to our negative findings, [30] finds distribution-free risk control overruns under group shift. We establish the corresponding speech anti-spoofing case and add attack-prevalence stress. ASVspoof 5 [18] audits a preset threshold in-domain, but its cross-dataset tables remain EER-only.

3. THRESHOLD POLICIES

A detector emits score s (higher = bona fide), and deployment flags spoof when $s < t$. We compare policies by resource class. With **no target data**, the source dev threshold t at the target FPR is transferred unchanged. With **unlabeled target data**, we fit the corrections of [7] to target statistics; we reimplement C1 z-norm, C2 temperature/shift, and C5 AS-norm. With N **labeled target bona-fide recordings**, we test three policies: (a) cohort z-norm standardizes scores by the cohort mean and standard deviation, then applies the source threshold in standardized units; (b) the Gaussian parametric quantile sets $t = \bar{s}_N - 1.645\,\hat{\sigma}_N$; and (c) the *conformal quantile* sets t to the k-th smallest cohort score, where $k = \lfloor (N{+}1)\alpha \rfloor$. Exchangeability without ties gives the last policy's marginal rank guarantee. Under iid continuous sampling from a fixed score distribution, its conditional FPR across calibration draws additionally follows $\mathrm{Beta}(k, N{+}1{-}k)$ [12, 13]. The **target-bona-fide oracle** sets the threshold realizing α on deployment bona fide; spoof labels measure its FNR but do not select it. All labeled policies consume identical paired cohort draws. **Definitions.** A detector–corpus cell is *overlap-dominated* when its oracle FNR at the target FPR exceeds 50%. Such cells are italicised in Table 1 and excluded from "viable" claims. Mean-control claims average calibration draws. The $\mathrm{Beta}(k, N{+}1{-}k)$ dispersion is an iid-continuous population reference, not the exact law of finite-pool draws without replacement. We report their held-out empirical spread separately over B=1000 draws. C4 (CORAL) of [7] requires embeddings through the frozen head, which is a different resource class, so C2 substitutes.

4. EXPERIMENTS

Setup. We evaluate AASIST [31], SSL-AASIST [5], and XLS-R+SLS [6]. The first two use released 19LA-trained checkpoints; where official XLS-R+SLS scores are available, our scores agree at per-trial r=0.994. The corpora are ASVspoof 2021 LA and DF [2], In-the-Wild [3], BRSpeech-DF [4], and ASVspoof 5 eval [18] for replication. We subsample DF to 100k trials for the reproduction-scored detectors and use the full 611k trials for official scores. Six of the seven 21LA conditions are real transmissions: five through an Asterisk PBX over VoIP and one over a PSTN carrier [2]. By contrast, 21DF is compression-only and untransmitted. All thresholds target α=5%. Labeled policies use the same paired cohort draws (B=1000), and FPR is measured on held-out data. Budget sweeps cover N=30–10^4; full-grid comparisons use N=500.

FPR control and FNR cost. At N=500, the conformal quantile's mean realized FPR is 4.94–5.03% across the 12 matched detector–corpus cells (Table 1). The budget sweep covers all eight cells. On ITW with SSL-AASIST, the empirical spread approaches the iid Beta reference; cohort z-norm remains 7–8 pp high at every budget, and the Gaussian quantile remains 0.6–1.6 pp high. On every viable cell, the quantile's FNR premium over the oracle threshold is ≤0.8 points.

Matched-resource policy comparison. With SSL-AASIST, N=500, and 1000 paired draws, the conformal quantile realizes 5.0% FPR on all four corpora, as expected from marginal rank validity. The informative comparison is how the alternatives behave with the same resources. Naive transfer misses target by 25–95 pp, and the best unlabeled correction misses by 3–95 pp. C5 AS-norm is excluded as degenerate because it controls FPR only by driving FNR to ≈99%. The Gaussian parametric quantile is the fair competitor at the same N because it also targets a target-domain FPR. It misses by up to 6.4 pp (0.66 pp on ITW) and collapses to 0% on BRSpeech. Cohort z-norm does not target a target-domain FPR, so its larger errors (3–18 pp, 6/8 cells beyond ±2 pp, and up to 50.8 pp for XLS-R+SLS) are a category difference rather than a direct contest. At N=100, its miss reaches 6.8 pp. Across the unlabeled corrections of [7], the miss is 3–95 pp.

Drift is detector-conditioned (Fig. 1). We measure severity as $\log_2(\mathrm{FPR}/\alpha)$. An additive band has only α points below target and therefore cannot distinguish a collapsed threshold from a controlled one, although the rankings are otherwise unchanged. On the ratio scale, 34 of the 84 cells passing our pre-registered ±5 pp test (a different 84 from the within-corpus set above, overlapping it on 68) miss target by over 2×, necessarily conservatively, since that band excludes a liberal 2× miss by construction. The 2× bar is post-hoc, and the count depends on that choice: it ranges from 60 of 84 at 1.5× to 22 of 84 at 4×. We therefore attach no interval. 68 of those 84 band-passing cells are within-corpus, where calibration and deployment are the same 2,636 source recordings from the same 67 speakers under seven 21LA channel conditions, so they are not independent observations and no significance claim is made of them. The replicate below tests whether the finding survives this dependence. Their spoof-side cost is measured against the oracle threshold for that deployment, not against the calibration condition, and reaches +0.55 (AASIST, PSTN→G.722; two cells score marginally higher but rest on too few deployment utterances to quote). Per calibration draw a positive spoof-side cost places the threshold below oracle and so inside the ±5 pp band; empirically every positive-price cell passes it, which is why the additive test cannot reveal this cost. AASIST is the more fragile detector on both axes. **Replication on recording-disjoint data.** Because the grid above reorders one 2,636-recording set, we repeat the experiment on ASVspoof 5 eval [18], where calibration and deployment share neither recording nor speaker. The grid contains 11 organizer-applied codecs plus the unprocessed source and 737 speakers. It is restricted to the 80.5% of bona-fide sources appearing under one condition and split 50/50 by speaker. **Over both detectors the transported threshold misses target by over 2× on 187 of 264 pairs (71%), against 50% on the grid above**: the effect persists without shared recordings or speakers; the numerical rate difference is confounded by corpus, codecs and attacks. The 19.5% of sources that *do* recur under every condition reproduce this corpus's twin structure and give 145 of 220 (66%), so twinning does not inflate severity in this corpus. This is descriptive evidence from one corpus, not a correction factor. **The spoof-side replication is narrower.** AASIST is overlap-dominated on this corpus (oracle FNR 63.9% against SSL-AASIST's 7.9%), so our own viability rule excludes it, and the flagship cell above is an AASIST cell. The mechanism replicates; that cell does not; the spoof-side axis here rests on one detector. **Exploratory cost-map sensitivity.** A fixed severity→price line fitted on 21LA gives held-out R^2 0.59 on 66 viable A5 SSL-AASIST cells (Spearman −0.90); deleting one A5 condition at a time gives R^2 = 0.45–0.64. We report these points descriptively, not as established inference: directed cells reuse 12 conditions and one detector, and no valid crossed-condition interval is available. Detector transfer also failed for the tested pair: an envelope fitted on one covers only 38–76% of the other's held-out cells against a 95% requirement.

Monitoring was a pre-registered co-primary negative-result analysis. The entropy monitor fails outright: its pre-registered Spearman correlations are −0.07/+0.07 and the corrected values are −0.17/−0.31. CDTS's entropy statistic reaches ROC-AUC 0.99 in the visual domain [32], and its v2 reports that miscalibration tracks competence at $r \approx -0.98$. Our severity–competence ρ of 0.52/0.54 has the same direction and is distinct from Fig. 1's W_1 correlation; we therefore claim no general dissociation. The mixture-W_1 monitor met its pre-registered criteria on SSL-AASIST (Spearman 0.64; TPR 0.92 at FPR 0.19), but that reading inherits the additive target: under the two-sided criterion it retains no operating point on either detector, its threshold was selected in-sample, and re-mixing deployment spoof share by ±50% pushes every benign cell over it in the −50% direction (30/31 and 17/19 at +50%). We report both readings and treat unlabeled drift monitoring as open. **Label-free heuristic.** We test an importance-weighted quantile that applies a clipped 15-bin ratio of deployment to calibration *mixture-score* densities to the bona-fide cohort. It holds 33/108 cells within 2 pp of target, compared with 26/108 unweighted. It improves 70 cells but restores a deployable operating point on neither detector. This is a heuristic rather than the construction of [27], which requires a target/source bona-fide covariate ratio and includes

Table 1. FNR (%) at the quantile threshold targeting mean FPR$\approx$5% vs oracle-threshold FNR (Δ). N=500, first calibration run; an independent re-draw moves no entry by more than 0.13 points. The manifest-bound supplement contains the exact PDF, source, guards, code and result JSONs.

System	21LA	21DF	ITW	BRSpeech
AASIST	36.4 (+0.8)	43.5 (+0.4)	*95.2 (0.0)*	*84.0 (−0.2)*
SSL-AASIST	15.9 (+0.7)	10.5 (+0.1)	18.4 (+0.3)	*98.5 (0.0)*
XLS-R+SLS†	1.6 (+0.1)	0.8 (+0.0)	11.2 (+0.4)	*91.5 (+0.4)*

†Official author-released scores (21DF full eval); italics = overlap-dominated cells where FPR control holds but no usable operating point exists.

the test-point weight; our negative result is limited to the heuristic. It also differs from the degeneracy in [33], where a high-dimensional logistic ratio clips every weight to zero; ours is one-dimensional and none reaches zero. Weight localisation is present and predicts the miss's *size* (ρ=-0.65) but not its *direction* (ρ=-0.07), and the conservative tendency localisation is predicted to induce does not appear (54/108 cells). Adaptive conformal [28] needs to know whether each flagged item was bona fide. A deployment with no target labels sees only the mixture flag rate, and recovering the false-alarm rate from it needs both the attack prevalence and the spoof-side detection rate. Mixture-proportion estimation cannot supply the first, because it requires the bona-fide score density in the deployment channel, which is the quantity this section shows moves. We therefore report it *with* oracle labels, as the ceiling labels buy (99/108 within 2 pp), and test no label-free form of it.

5. DISCUSSION

Calibration and detection are distinct. Target-channel calibration controls the *marginal mean*, not every realized threshold. Under iid continuous sampling from a fixed distribution, a single N=100 draw has only 65.5% probability of landing within $\pm$2 FPR points; at N=500, the probability is 96.2%, with a 3.26–7.06% central 95% interval. Calibration also cannot repair poor discrimination: on overlap-dominated cells, no threshold yields a deployable operating point [7]. **Operational implications.** Report FNR at controlled FPR alongside EER, choose N from the dispersion required by the deployment, ship a designated bona-fide calibration split, and recalibrate after a channel change. **Limitations.** The Beta dispersion law assumes iid continuous recordings from a fixed deployment speaker mixture. A post-hoc speaker-disjoint diagnostic on 21LA widened the estimated 90% FPR interval for AASIST/PSTN to 7.03 points (MC SD 0.17), compared with a speaker-permutation null near 3.4; deployment to new speakers therefore requires speaker-level validation. Conclusions rest on three detectors, the drift map on the two reproduction-scored ones and the ASVspoof 5 replicate on those two; our language-shift regime rests on

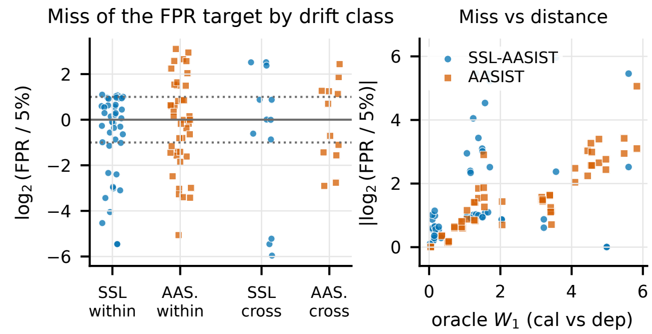

Fig. 1. Drift map. Left: signed $\log_2(\mathrm{FPR}/5\%)$ per deployment cell (within-21LA channel pairs vs cross-corpus; colour/marker = detector). Under a post-hoc $2\times$ two-sided bar, 34 of the 84 band-passing cells are off target (60 at $1.5\times$, 22 at $4\times$; no interval, §4), every one conservative. The costliest on the spoof side are among them: AASIST PSTN$\rightarrow$G.722 misses 74% of spoofs against 19% for that deployment's oracle while reading 0.47% FPR. A resolution-limited SSL GSM$\rightarrow$none cell combines an AUC rise, 0.977$\rightarrow$0.982, with a rule-of-three-censored 44$\times$ FPR miss: an existence proof, not a rate. Right: the miss correlates with oracle bona-fide W_1 (ρ 0.52/0.88) but does not transfer across detectors (held-out R^2 0.48/-0.02), so no predictive law is claimed from it. Cells are not independent replications; within-corpus slices reuse recordings and speakers (§4).

one corpus (n=1), [11] covering 66 but on the spoof side only; and bona-fide labeling is assumed cheap relative to spoof labeling, which holds where a deployment can verify its own recordings' provenance. The within-corpus slices are repeated views of one sample; the recording- and speaker-disjoint replicate tests the transport finding without that dependence. Price and severity are two readouts of threshold displacement. Their strong within-grid rank association does not make spoof-side cost identifiable from bona-fide scores alone.

6. CONCLUSION

Threshold transport is easy to mis-audit: conservative collapse can make FPR look better while missed spoofs quadruple. A bona-fide order statistic from the target channel restores marginal expected control, but dispersion depends on sample size and sampling unit, and cannot repair class overlap. Recording- and speaker-disjoint data confirm transport failure; the label-free heuristic tested does not restore control.

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